

Model Development Phase Template

Date	02 Feb 2026
Team ID	LTVIP2026TMIDS75146
Project Title	Rising Waters: A Machine Learning Approach To Flood Prediction
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code will be showcased in the future through a screenshot. The model validation and evaluation report will include classification reports, accuracy, and confusion matrices for multiple models, presented through respective screenshots.

Initial Model Training Code:

```
# importing and building the Decision Tree model
dtree = tree.DecisionTreeClassifier()
dtree.fit(X_train_scaled,y_train)

p_1=dtree.predict(X_train_scaled)
print("Training accuracy through the decision tree method is:", accuracy_score(y_train, p_1))
p1 = dtree.predict(X_test_scaled)
print("The accuracy through the decision tree method is:", accuracy_score(y_test, p1))
```

```
# importing and building the Random Forest model
Rf = ensemble.RandomForestClassifier()
Rf.fit(X_train_scaled,y_train)
p_2=dtree.predict(X_train_scaled)
print("Training accuracy through the random forest method is:", accuracy_score(y_train, p_2))
p2 = Rf.predict(X_test_scaled)
print("The accuracy through the random forest method is:",accuracy_score(y_test, p2))
```

```
# importing and building the Knn model
knn = neighbors.KNeighborsClassifier()
knn.fit(X_train_scaled,y_train)
p_3=knn.predict(X_train_scaled)
print("Training accuracy through the Knn method is:",accuracy_score(y_train, p_3))
p3 = knn.predict(X_test_scaled)
print("The accuracy through the random forest method is:",accuracy_score(y_test, p3))
```

```
# importing and building the Xgb model
xgb = xgboost.XGBClassifier()
xgb.fit(X_train_scaled,y_train)
p_4=xgb.predict(X_train_scaled)
print("Training accuracy through the Xgb method is:",accuracy_score(y_train, p_3))
p4 = xgb.predict(X_test_scaled)
print("The accuracy through the Xgb method is:",accuracy_score(y_test, p3))
```

Model Validation and Evaluation Report:

Model Validation and Evaluation Report								
Model	Classification Report				Accuracy	Confusion Matrix		
Decision Tree	<pre>print(classification_report(y_test,p1)) print(confusion_matrix(y_test,p1))</pre>				100%	[[20 0] [0 3]]		
		precision	recall	f1-score			support	
	0	1.00	1.00	1.00			20	
	1	1.00	1.00	1.00			3	
		accuracy		1.00			23	
		macro avg	1.00	1.00			1.00	23
		weighted avg	1.00	1.00			1.00	23
Random Forest	<pre>print(classification_report(y_test,p2)) print(confusion_matrix(y_test,p2))</pre>				100%	[[20 0] [0 3]]		
		precision	recall	f1-score			support	
	0	1.00	1.00	1.00			20	
	1	1.00	1.00	1.00			3	
		accuracy		1.00			23	
		macro avg	1.00	1.00			1.00	23
		weighted avg	1.00	1.00			1.00	23

KNN	<pre>print(classification_report(y_test,p3)) print(confusion_matrix(y_test,p3))</pre> <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>0</td><td>0.91</td><td>1.00</td><td>0.95</td><td>20</td></tr><tr><td>1</td><td>1.00</td><td>0.33</td><td>0.50</td><td>3</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.91</td><td>23</td></tr><tr><td>macro avg</td><td>0.95</td><td>0.67</td><td>0.73</td><td>23</td></tr><tr><td>weighted avg</td><td>0.92</td><td>0.91</td><td>0.89</td><td>23</td></tr></tbody></table>		precision	recall	f1-score	support	0	0.91	1.00	0.95	20	1	1.00	0.33	0.50	3	accuracy			0.91	23	macro avg	0.95	0.67	0.73	23	weighted avg	0.92	0.91	0.89	23	91.3%	<pre>[[20 0] [2 1]]</pre>
	precision	recall	f1-score	support																													
0	0.91	1.00	0.95	20																													
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macro avg	0.95	0.67	0.73	23																													
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XGB	<pre>print(classification_report(y_test,p4)) print(confusion_matrix(y_test,p4))</pre> <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>0</td><td>1.00</td><td>1.00</td><td>1.00</td><td>20</td></tr><tr><td>1</td><td>1.00</td><td>1.00</td><td>1.00</td><td>3</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>23</td></tr><tr><td>macro avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>23</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>23</td></tr></tbody></table>		precision	recall	f1-score	support	0	1.00	1.00	1.00	20	1	1.00	1.00	1.00	3	accuracy			1.00	23	macro avg	1.00	1.00	1.00	23	weighted avg	1.00	1.00	1.00	23	91.3%	<pre>[[20 0] [0 3]]</pre>
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0	1.00	1.00	1.00	20																													
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